



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal to DEVI TRADERS

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

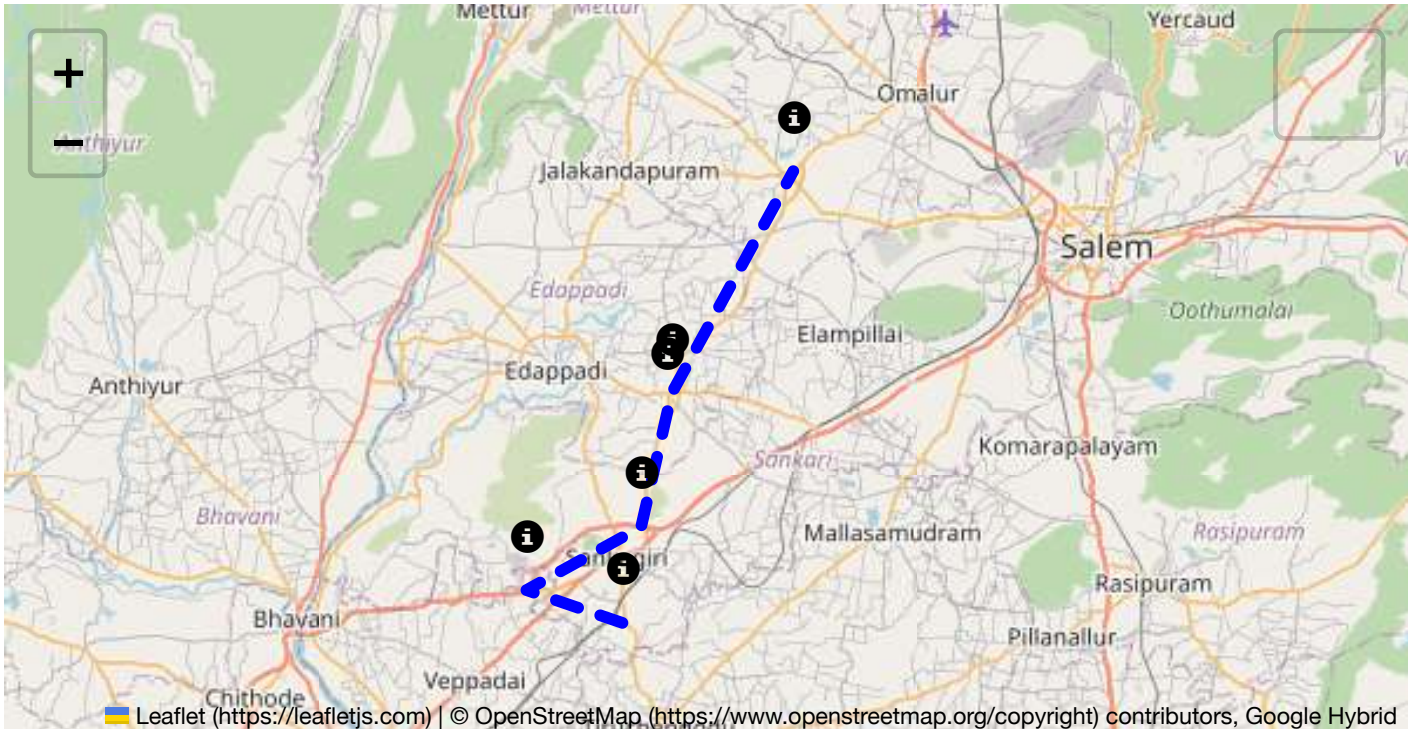
The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

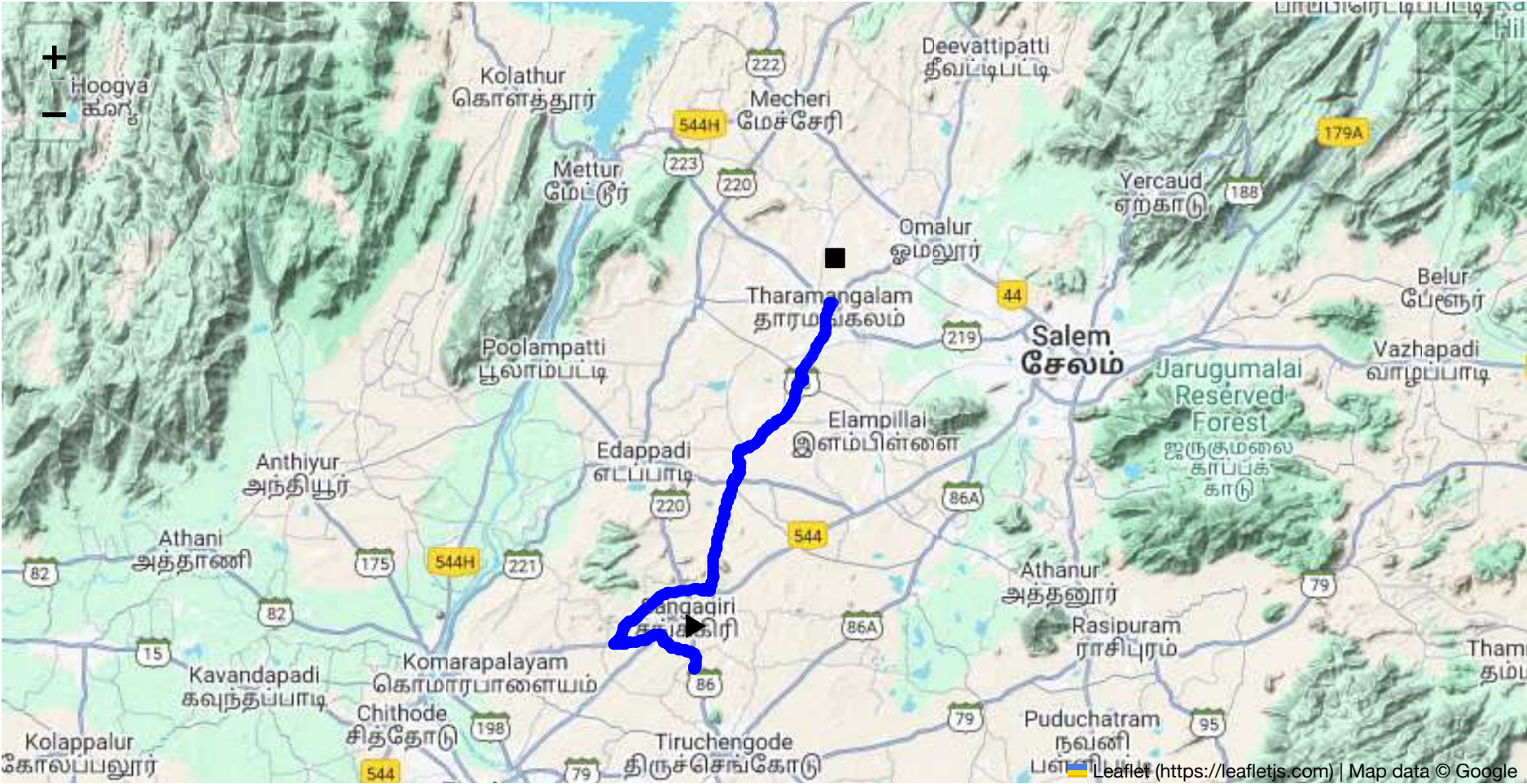
Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 44.83 km
Estimated Duration: 1.1 hours
Adjusted Duration (Heavy Vehicle): 1.4 hours

Start: (11.4381, 77.8734)
End: (11.697304, 77.973729)



Welcome to the Journey Risk Management Study

Route Overview

The route from CVQF+23W, Sangagiri to MXWF+XHQ, Tharamangalam spans approximately 44.83 kilometers. It travels through Bhavani Main Road in Padaiveedu, SH 86 in Ayveli, and Erumaipatti Road, before reaching its final destination in Tharamangalam. The route is a mix of state highways and local roads, primarily used by both commercial and residential vehicles.

Typical Weather Conditions and Hazards

Tamil Nadu experiences a tropical climate with high temperatures and significant rainfall during monsoon seasons (June to September and October to December). Expect potential hazards related to heavy rains such as flooding or water-logging, especially on roads with poor drainage. Heat during the summer months (March to May) may lead to road surface deterioration.

Traffic Patterns

Traffic is generally moderate but can become congested during peak hours, typically between 8:00-10:00 AM and 5:00-8:00 PM, especially near larger towns and commercial areas. Be wary of congestion at intersections and marketplaces, particularly in urban areas along Bhavani Main Road and SH 86.

Road Quality and Infrastructure

The general road infrastructure is reasonable but varies along the route. State highways like SH 86 are in better condition, but local roads might present challenges such as potholes and uneven surfaces. Watch for local construction activities which can cause temporary roadblocks or detours.

Alternative Routes for Emergencies

In case of emergencies, consider using National Highway 44, which runs parallel to the given route, though it may increase travel distance. Stay updated on local detours that might offer quicker alternatives during peak congestion or road closures.

Local Regulations Affecting Hazardous Material Transport

Tamil Nadu requires specific permits for transporting hazardous materials. Ensure adherence to scheduled transport times, often restricted during high-traffic periods. Vehicles should also be compliant with environmental and safety standards as per local and national regulations.

Historical Incidents Involving Heavy Vehicles or Hazardous Materials

There have been reports of accidents involving heavy transport on state highways in Tamil Nadu, mostly due to high speed and violation of traffic norms. No specific recurring incidents have been noted, but caution is advised in accident-prone zones marked by local authorities.

Environmental Considerations and Sensitive Areas

The route passes through agricultural and semi-residential zones. Care should be taken to prevent spills of hazardous materials that could affect local ecosystems. Some areas may be near protected natural habitats; adhere strictly to environmental regulations to mitigate risks.

Communication Coverage

Mobile network coverage is generally good along state highways and main roads. However, some stretches, especially through remote areas or between towns, may experience intermittent signal loss. Pre-plan communication needs and have alternative communication plans, such as satellite phones or vehicle-installed transceivers.

Estimated Emergency Response Times

Emergency response times may vary from 15 minutes in urbanized areas to up to 45 minutes in remote parts of the route. Proximity to nearby towns makes access easier for EMS facilities based in larger municipalities such as Sangagiri and Tharamangalam.

Overall Summary of Risk Assessment

The overall risk assessment for this route is moderate. Major risks stem from weather-related conditions and varying road quality. Traffic congestion during peak hours and compliance with local transport regulations are crucial points for effective route management. Preparing for communication gaps and potential detours will enhance safety and operational efficiency. Regular updates and awareness of changing conditions can significantly minimize potential hazards.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Roundabout	High	11.56180, 77.90019	15 KM/Hr
1	Roundabout	High	11.57053, 77.90246	15 KM/Hr
2	U-Turn	High	11.5729264, 77.9020974	10 KM/Hr
3	Turn	High	11.43811, 77.87348	15 KM/Hr
4	Turn	Medium	11.43956, 77.87340	30 KM/Hr
5	Turn	High	11.43968, 77.87345	15 KM/Hr
6	Turn	High	11.44029, 77.87544	15 KM/Hr
7	Turn	Medium	11.45353, 77.85697	30 KM/Hr
8	Turn	Medium	11.45839, 77.85143	30 KM/Hr
9	Turn	Medium	11.46307, 77.84941	30 KM/Hr
10	Turn	Medium	11.46316, 77.84921	30 KM/Hr
11	Turn	High	11.45542, 77.81725	15 KM/Hr
12	Blind Spot	Blind Spot	11.45631, 77.81668	10 KM/Hr
13	Turn	High	11.49409, 77.88004	15 KM/Hr
14	Turn	Medium	11.49420, 77.88016	30 KM/Hr
15	Turn	High	11.49382, 77.88530	15 KM/Hr
16	Turn	High	11.49408, 77.88538	15 KM/Hr
17	Turn	Medium	11.49444, 77.88492	30 KM/Hr
18	Turn	Medium	11.49469, 77.88488	30 KM/Hr
19	Turn	Medium	11.56096, 77.89931	30 KM/Hr
20	Blind Spot	Blind Spot	11.57236, 77.90352	10 KM/Hr
21	Blind Spot	Blind Spot	11.57293, 77.90210	10 KM/Hr
22	Blind Spot	Blind Spot	11.57236, 77.90352	10 KM/Hr
23	Turn	Medium	11.58304, 77.90656	30 KM/Hr
24	Turn	Medium	11.64197, 77.95197	30 KM/Hr
25	Turn	Medium	11.64721, 77.95118	30 KM/Hr
26	Turn	High	11.69590, 77.97114	15 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
1	hospital	Sri Shanthi Hospital	11.569272, 77.901937	30 km/h	Medium
2	hospital	Dr. Pandiyarajan Hospital	11.692704, 77.969344	30 km/h	Medium
3	clinic	Sun Eye Care	11.6936242, 77.9730948	30 km/h	Medium
4	hospital	Dr. Sengodan Hospital	11.6959527, 77.9695325	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
0	school	KRP Matric. Hr. Sec School	11.4546193, 77.8142445	30 km/h	Medium

Route Photos of Risky Spots



Risk Type: Roundabout

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.56180, 77.90019



Risk Type: Roundabout

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.57053, 77.90246



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.44029, 77.87544



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.45353, 77.85697



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.45839, 77.85143



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.46307, 77.84941



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.46316, 77.84921



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.45542, 77.81725



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.45631, 77.81668



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.49409, 77.88004



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.49420, 77.88016



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.49382, 77.88530



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.49408, 77.88538



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.49444, 77.88492



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.49469, 77.88488



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.56096, 77.89931



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.57236, 77.90352



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.57236, 77.90352



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.58304, 77.90656



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.64197, 77.95197



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.64721, 77.95118



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.69590, 77.97114

